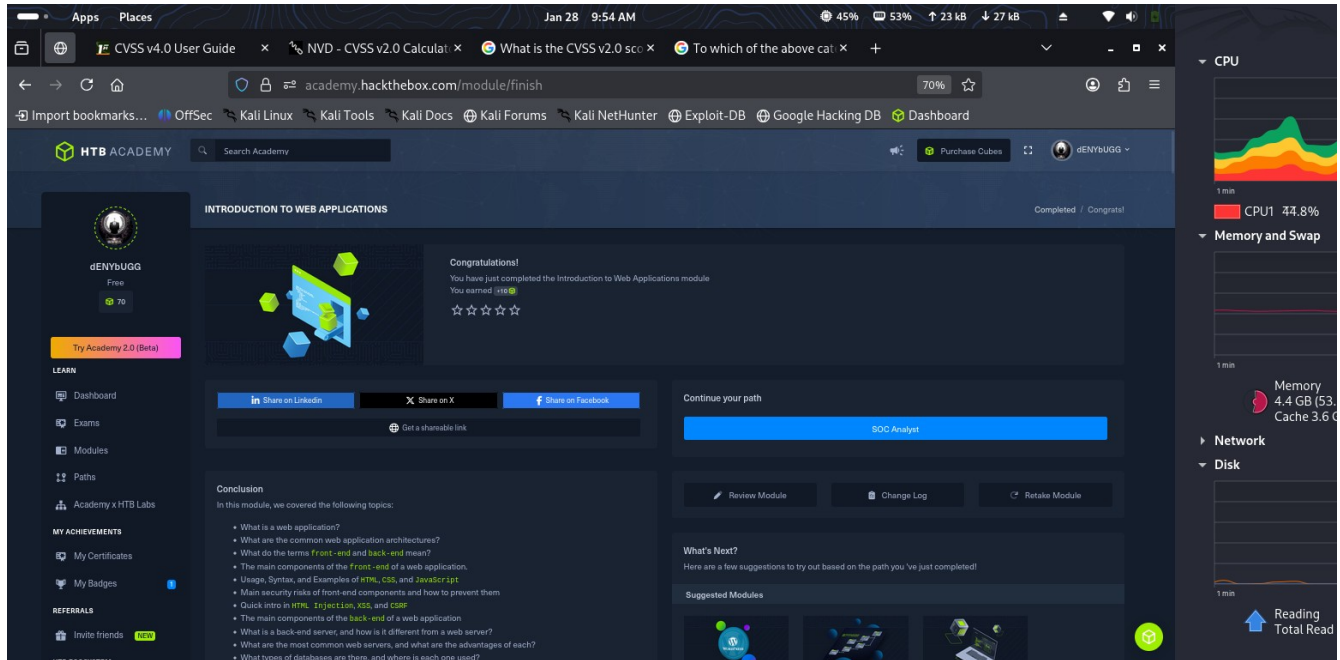


NAME: Alex Mbogo
GMAIL: evansnjiru5@gmail.com
ADMISSION NUMBER: ADC-CSS02-25064

INTRODUCTION TO WEB APPLICATIONS

Module Completion Link

<https://academy.hackthebox.com/achievement/1378833/75>



1:1 Introduction

Web applications are dynamic software programs that run in a browser, using a client-server architecture to process data and interact with users in real time.

1:2 Web Applications vs Websites

Websites held static information that would not change d with user interaction in real time web applications are fully functional and can perform various functionalities for the end-user, while web sites lack this type of functionality.

1:3 Attacking Web Applications

Attacks that can be carried out on web applications include:Sql injection, File inclusion , Unrestricted File Upload, Insecure Direct Object Referencing (IDOR) Broken Accesss Control

2:0 Web Application Layout

2:1 Web Application Infrastructure

- Client-Server
- One Server
- Many Servers - One Database
- Many Servers - Many Databases

there are other web application models available such as server less web applications or web applications that utilize micro services

2:2 Web Applications Components

- Client
 - Webserver
 - Web Application Logic
 - Database
- Services (Microservices)
 - 3rd Party Integrations
 - Web Application Integrations
- Functions (Serverless)

3.0 Front-end vs Back End

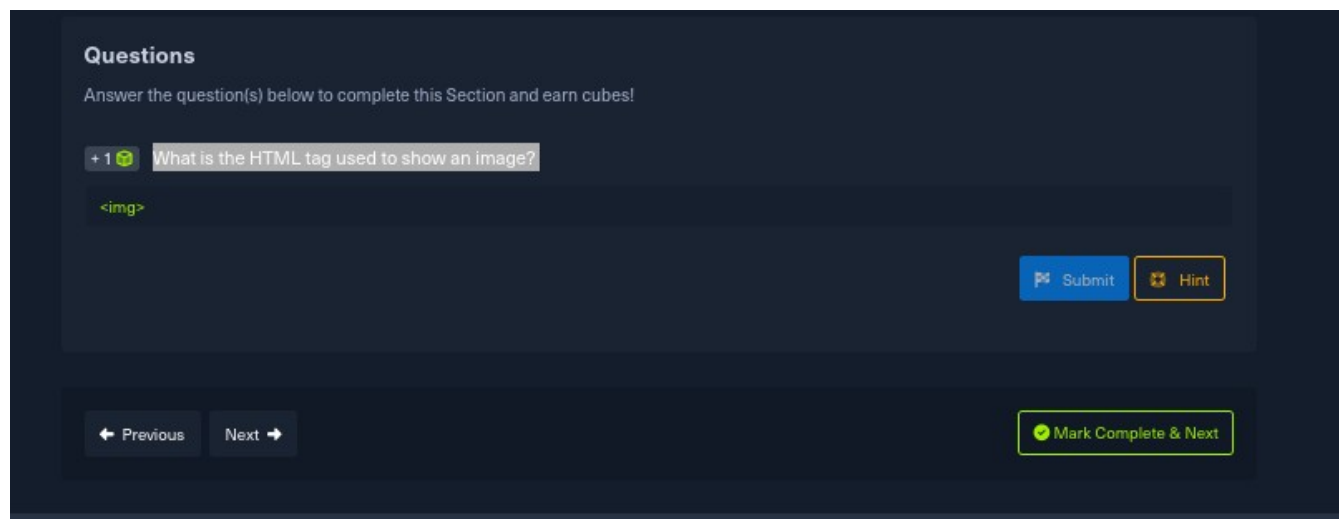
The front end of a web application contains the user's components directly through their web browser (client-side)

The back end of a web application drives all of the core web application functionalities, all of which is executed at the back end server, which processes everything required for the web application to run correctly

3:1 Front-end

HTML

What is the HTML tag used to show an image?

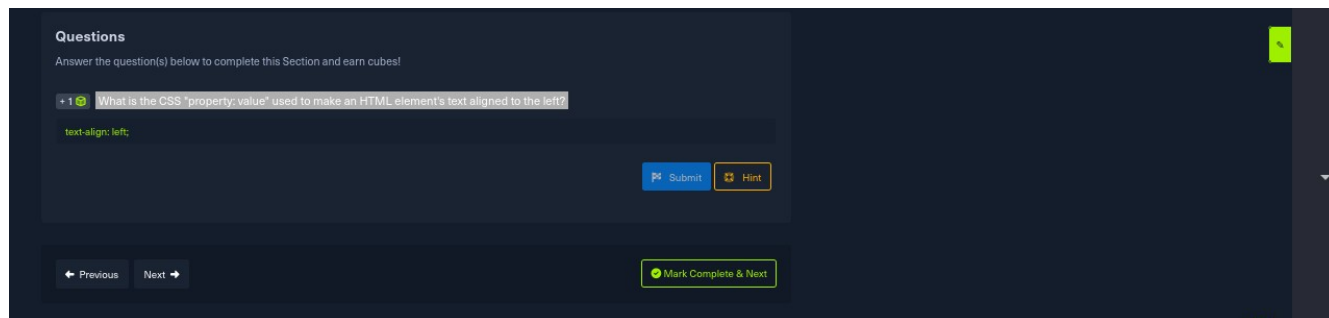


The screenshot shows a quiz interface with a dark blue background. At the top, the word "Questions" is displayed in white. Below it, a prompt says "Answer the question(s) below to complete this Section and earn cubes!". A question box contains the text "What is the HTML tag used to show an image?" with a "+1" icon and a green cube icon. Below the question box, the text "" is entered. To the right of the input field are two buttons: "Submit" (blue) and "Hint" (yellow). At the bottom of the interface, there are two navigation buttons: "Previous" (with a left arrow) and "Next" (with a right arrow). On the far right, there is a button labeled "Mark Complete & Next" with a green checkmark icon.

CSS

Css is the stylesheet language used alongside HTML to format and set the style of HTML elements.

What is the CSS "property: value" used to make an HTML element's text aligned to the left?



JavaScript

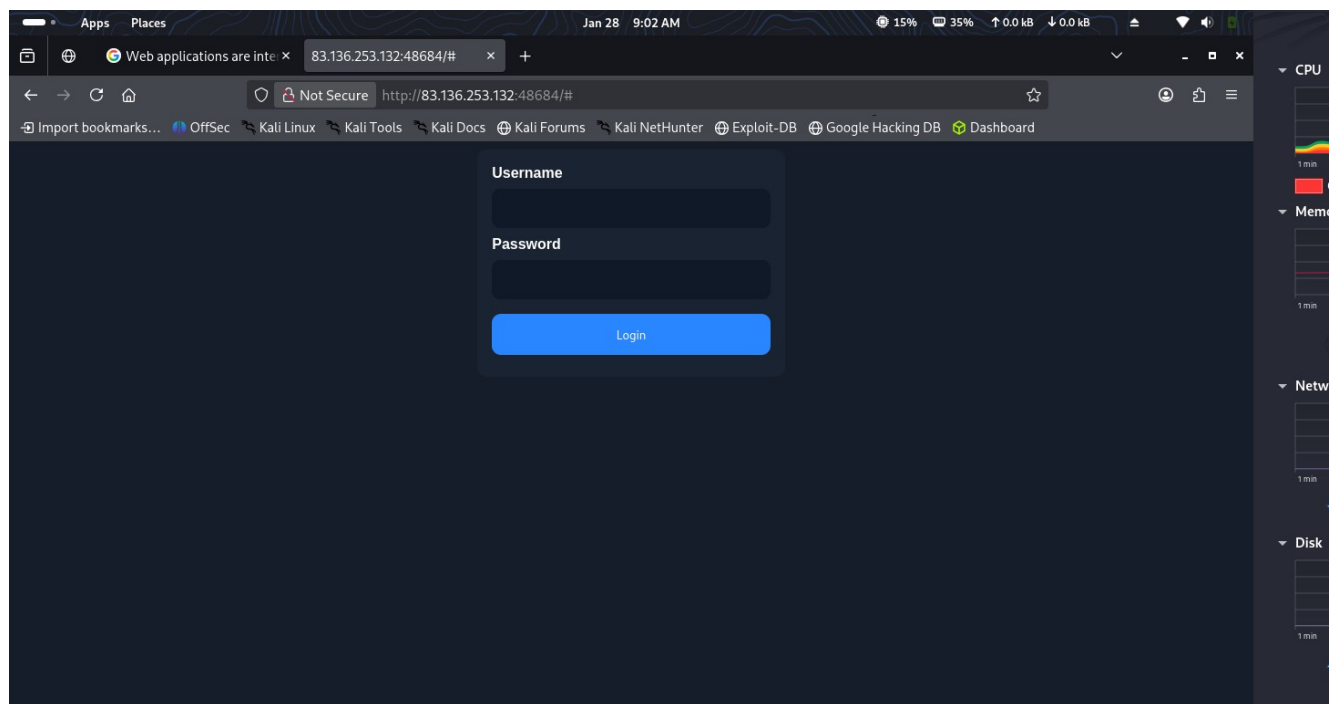
JavaScript is a programming language used to transform static web pages into interactive, dynamic applications. While it traditionally handles front-end behavior, modern tools like Node.js allow it to power the entire application stack.

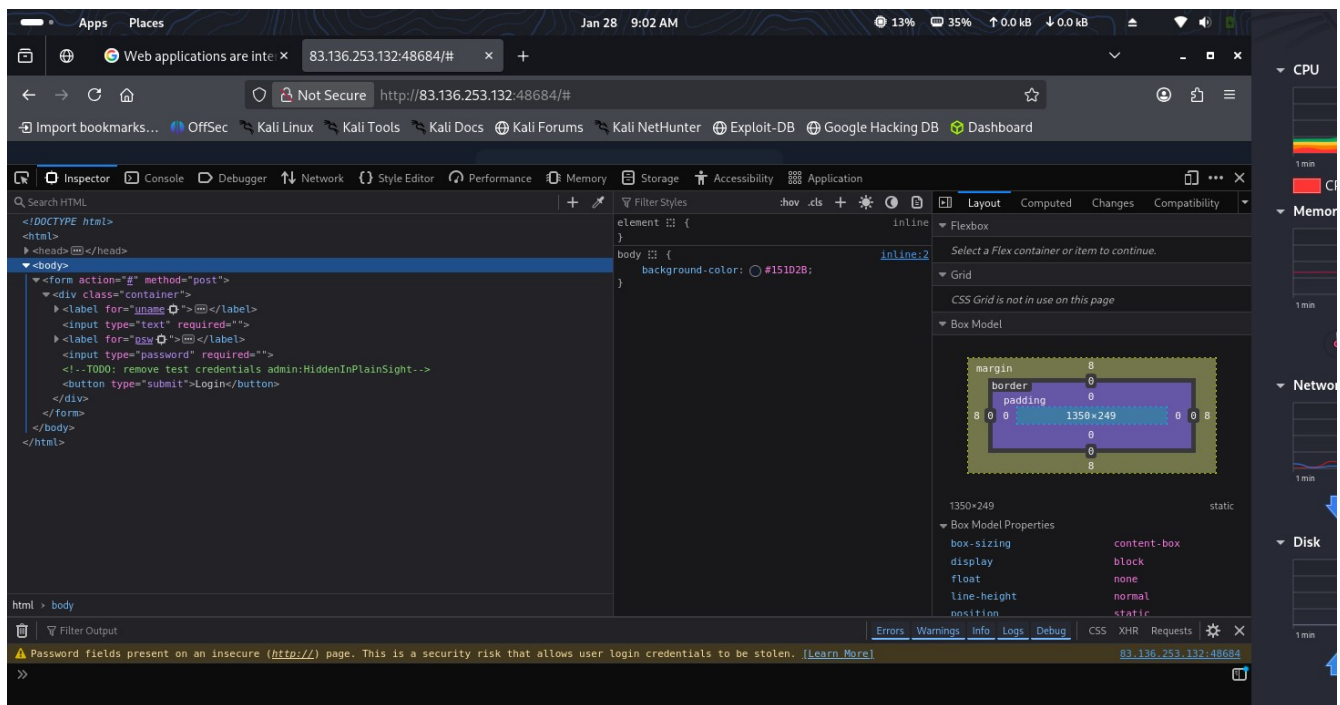
3.1.0 Front End Vulnerabilities

3:1:1 Sensitive Data Exposure

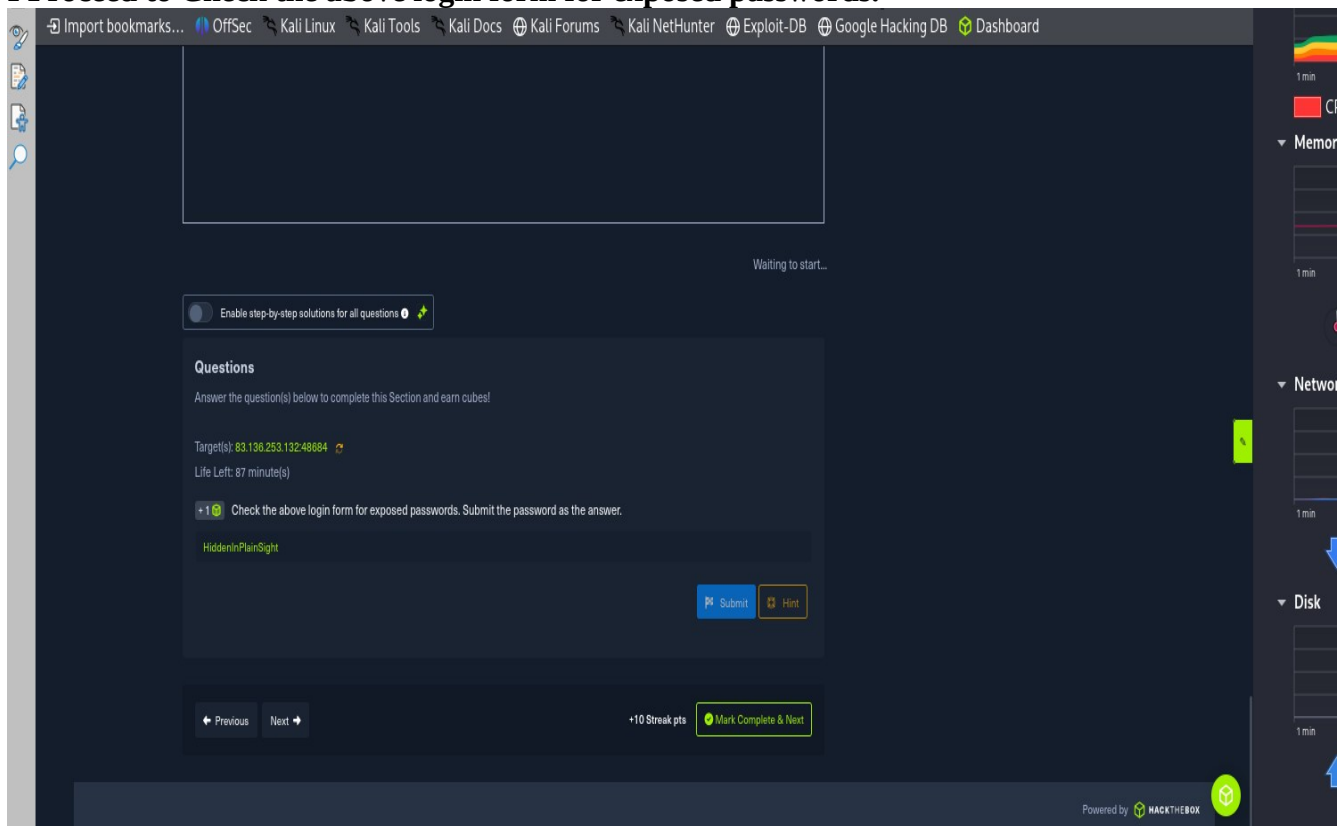
refers to the availability of sensitive data in clear-text to the end-user. This is usually found in the source code of the web page or page source on the front end of web applications.

Lets look for the password int the source code





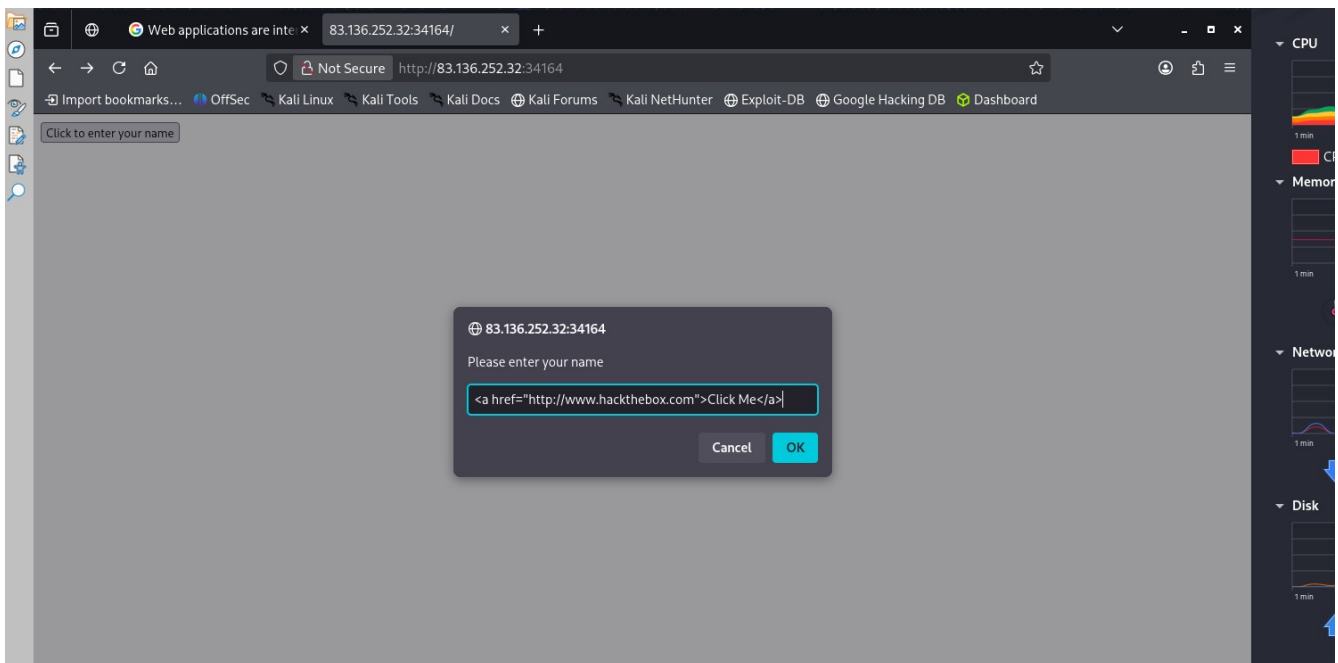
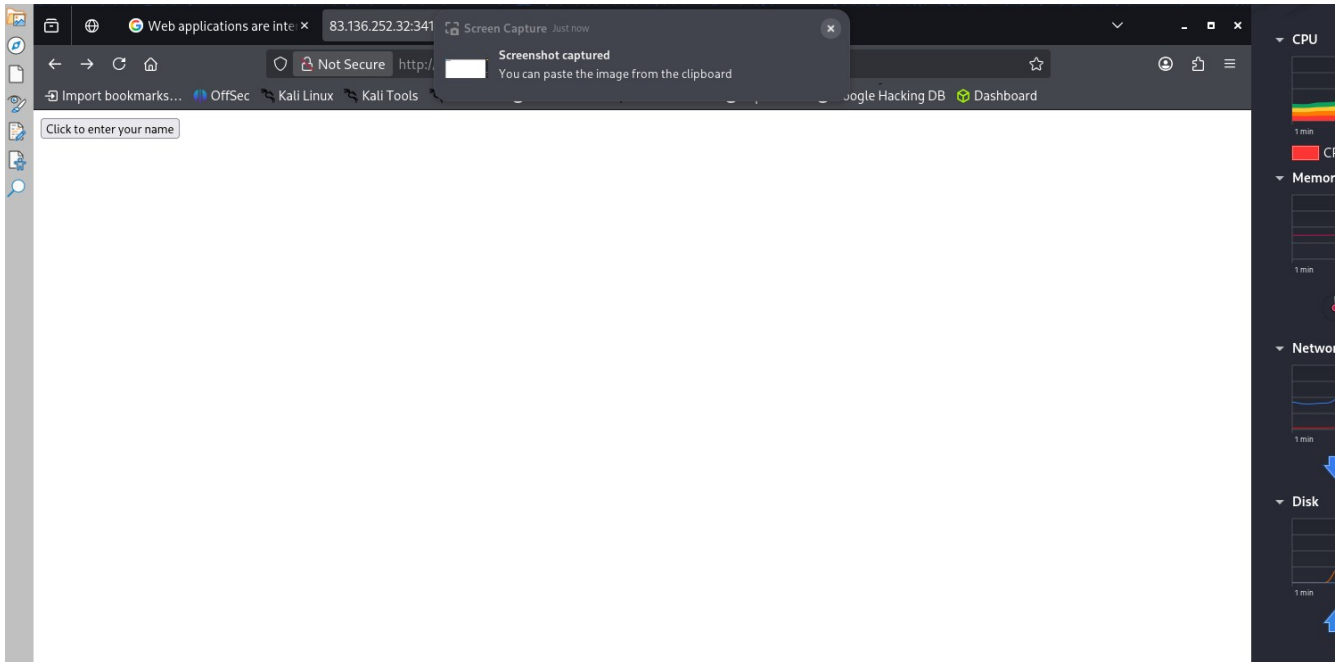
I Proceed to Check the above login form for exposed passwords.

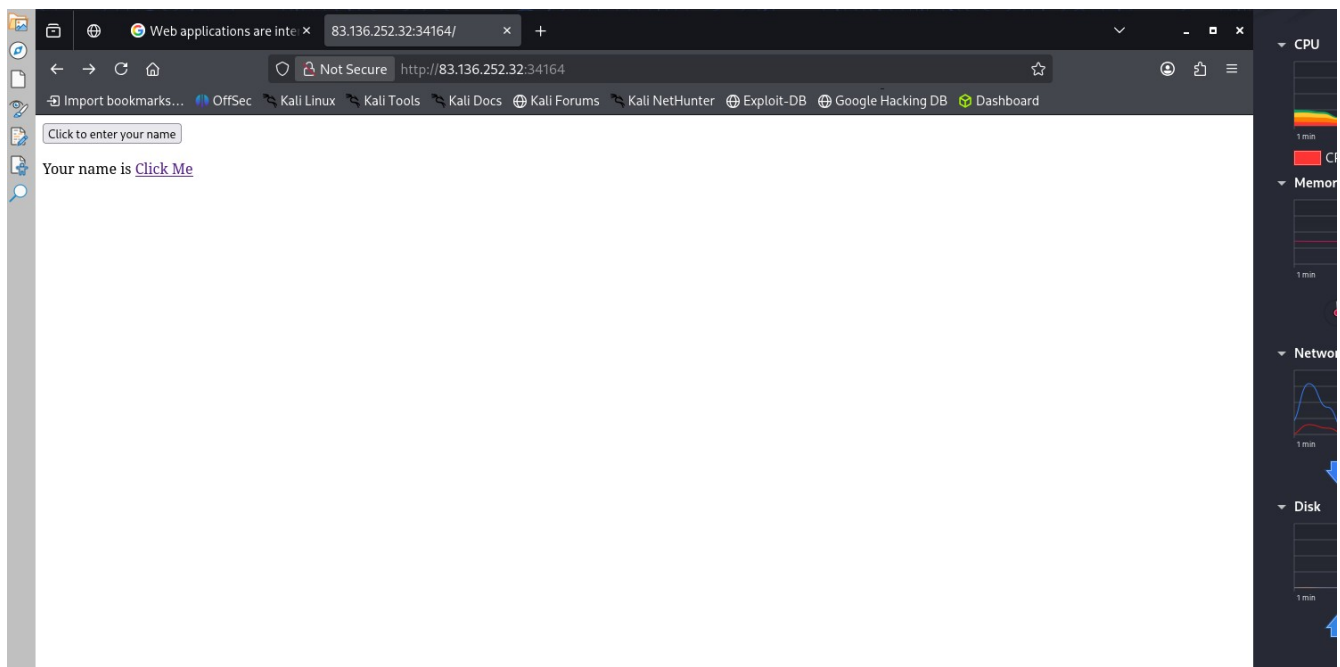


3:1:2 HTML Injection

HTML Injection occurs when unfiltered user input is displayed on the page.

What text would be displayed on the page if we use the following payload as our input: `<a href="http://www.hackthebox.com">Click Me</a>`



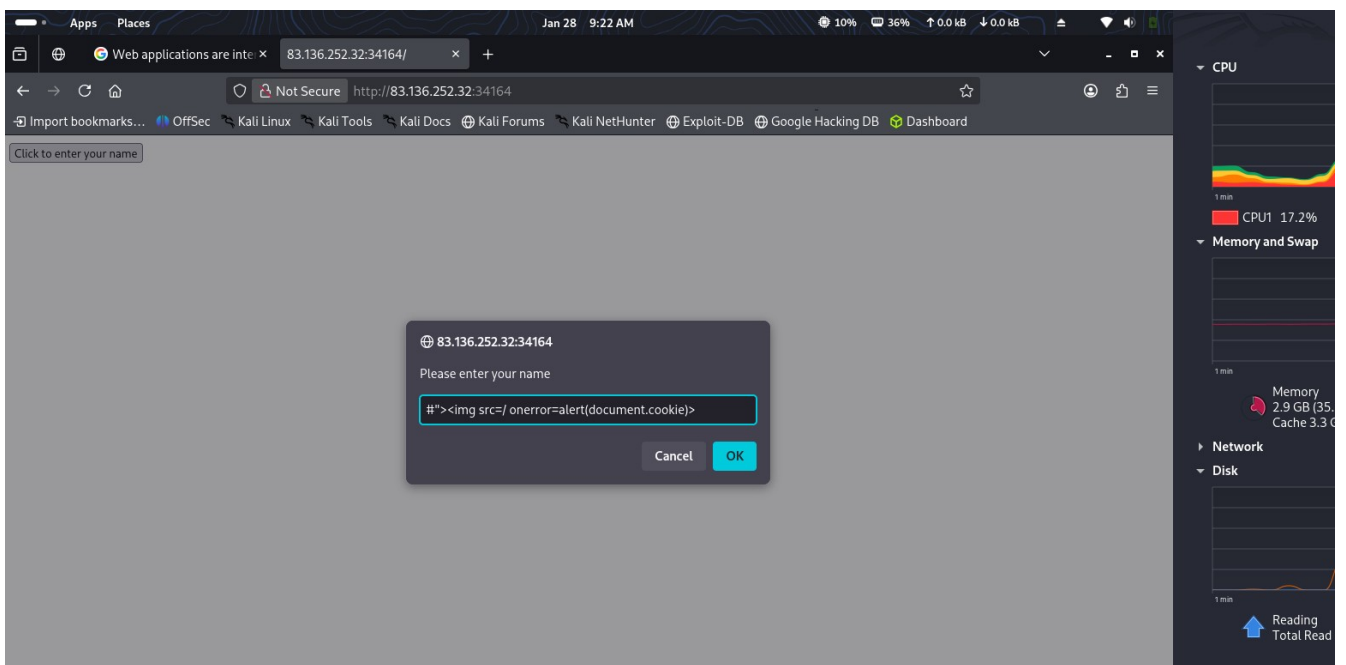
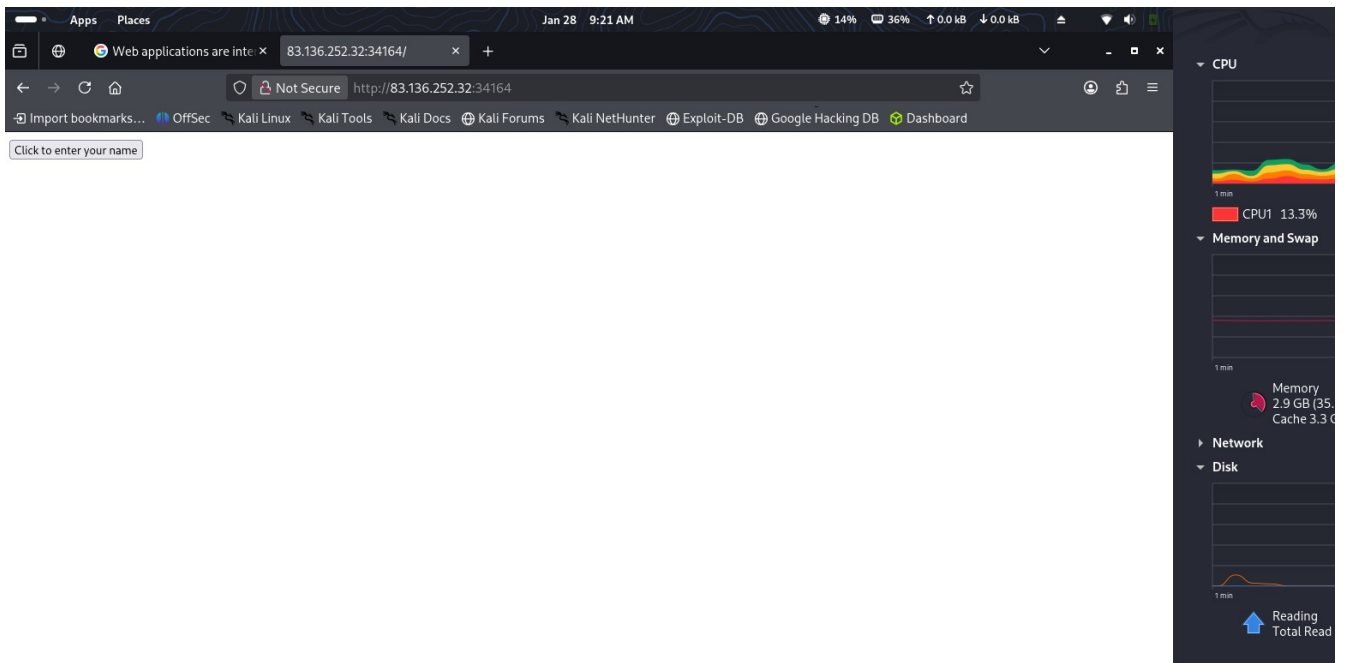


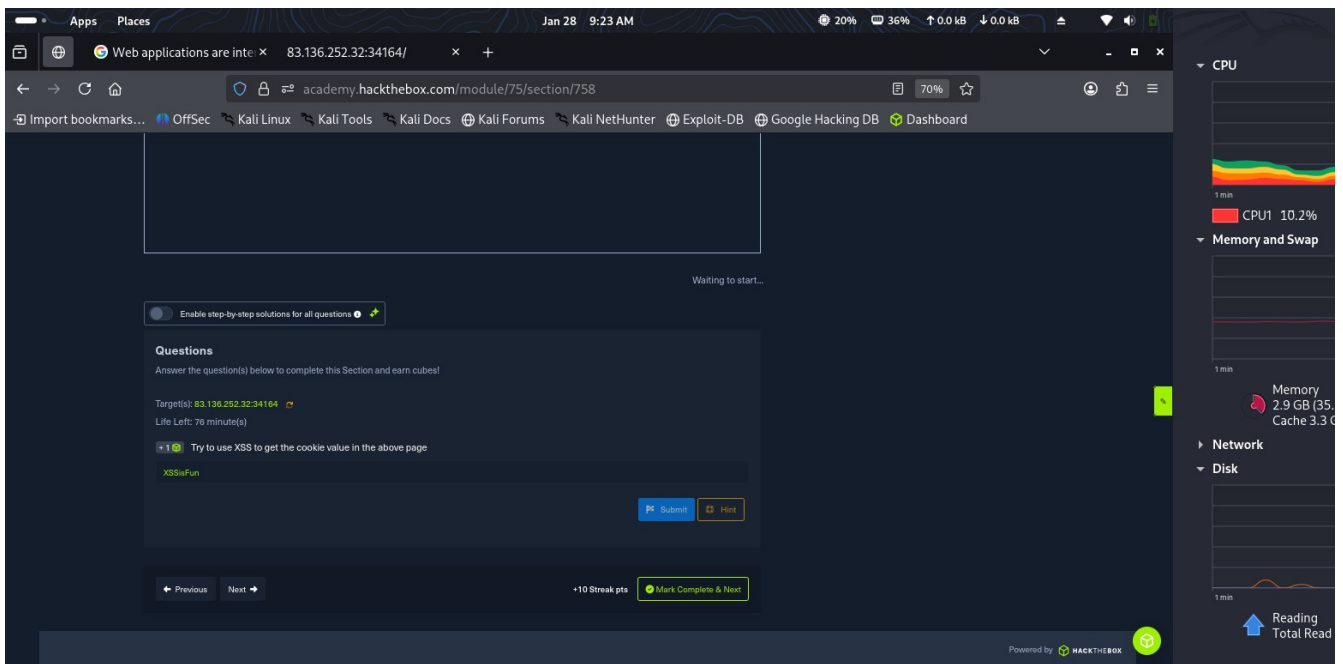
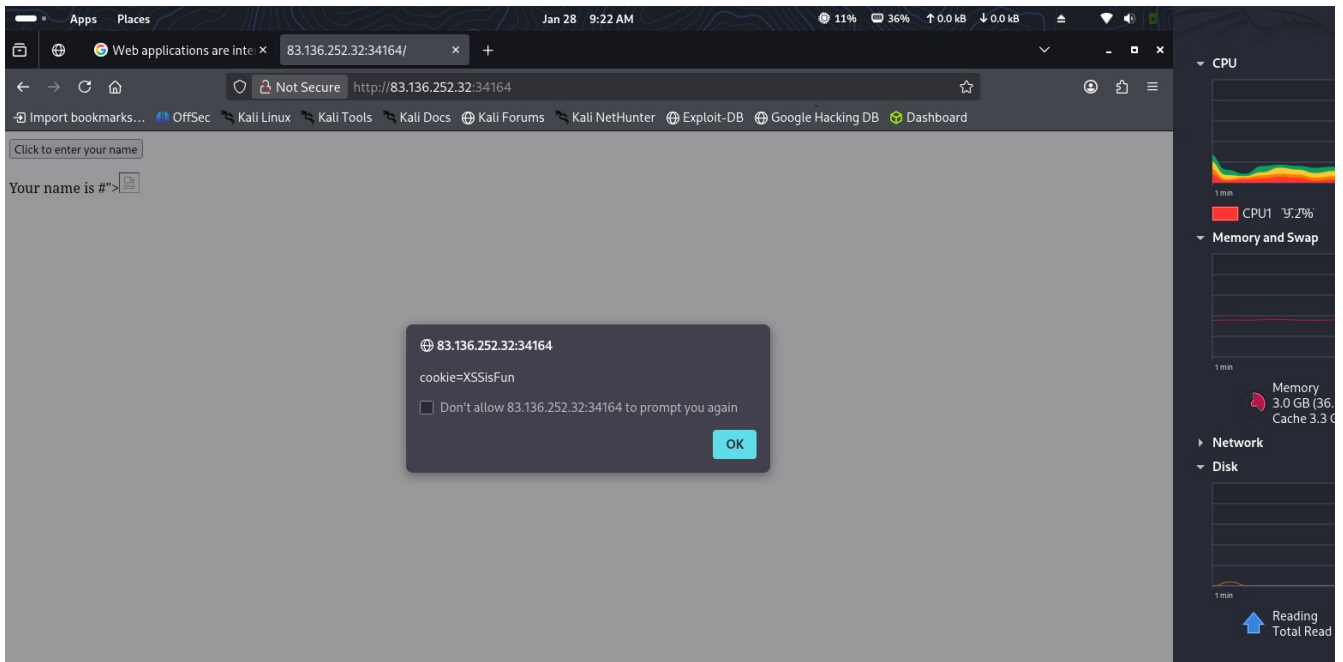
3:2:3 Cross-Site Scripting (XSS)

Vulnerabilities can often be utilized to also perform XSS attacks by injecting JavaScript code to be executed on the client-side. Once we can execute code on the victim's machine, we can potentially gain access to the victim's account or even their machine.

- Reflected XSS Occurs when user input is displayed on the page after processing (e.g., search result or error message).
- Stored XSS Occurs when user input is stored in the back end database and then displayed upon retrieval (e.g., posts or comments).
- DOM XSS Occurs when user input is directly shown in the browser and is written to an HTML DOM object (e.g., vulnerable username or page title).

Performing XSS on A target





3:2 Back End Components

3:2:1 Back End Servers

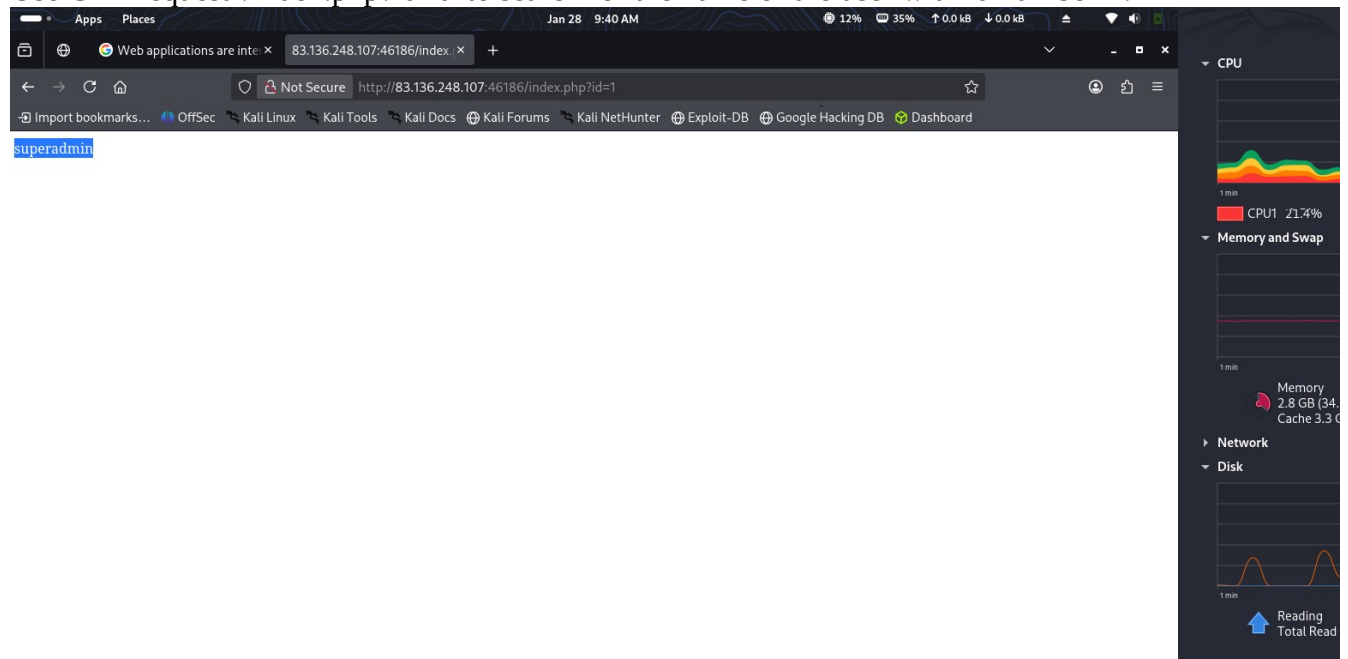
A back-end server is the hardware and operating system on the back end that hosts all of the applications necessary to run the web application. It is the real system running all of the processes and carrying out all of the tasks that make up the entire web application.

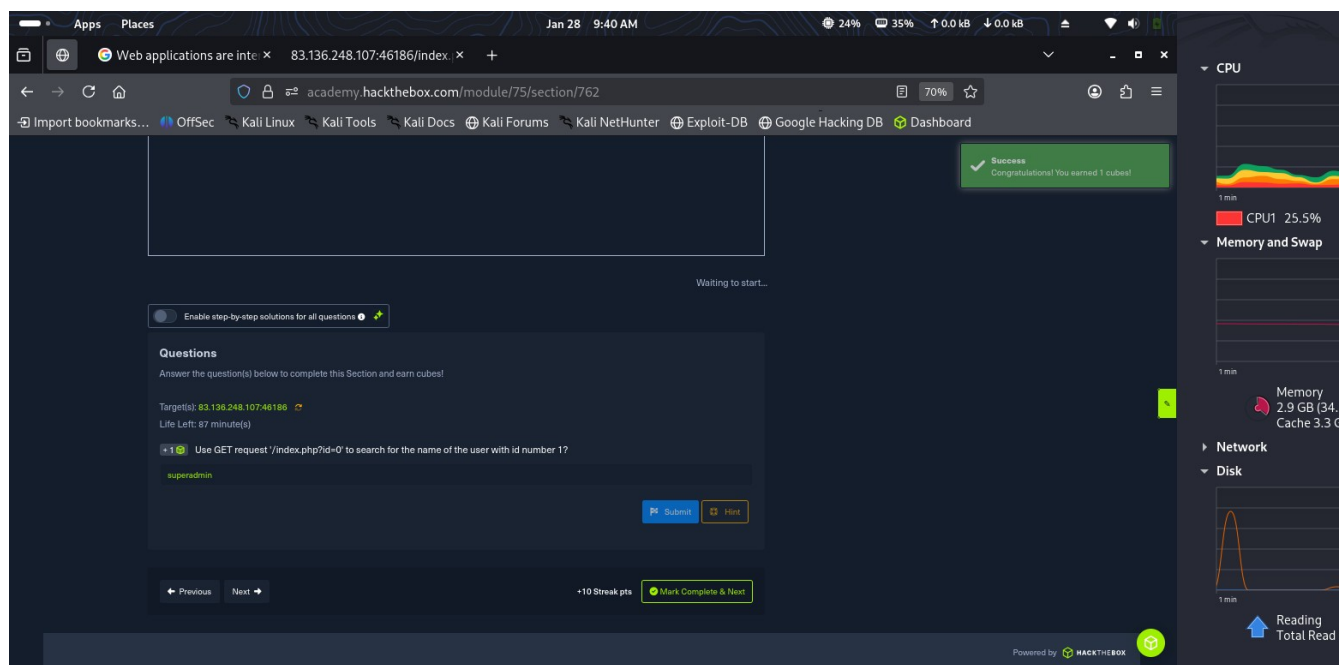
3:2:2 Web Servers

an application that runs on the back end server, which handles all of the HTTP traffic from the client-side browser, routes it to the requested pages, and finally responds to the client-side browser. Web servers usually run on TCP ports 80 or 443, and are responsible for connecting end-users to various parts of the web application, in addition to handling their various responses.

3:3 Development Frameworks & APIs

Use GET request '/index.php?id=0' to search for the name of the user with id number 1?





4:0 Common Web Vulnerabilities

Broken Authentication/Access Control

Malicious File Upload

Command Injection

SQL Injection (SQLi)

Public Vulnerabilities

Public CVE